

MAT 017B Quiz 1

August 7, 2018

Name: Key

SID: _____

1. Find a particular antiderivative of the following functions.

a. $5e^{-x}$

$$\left(\frac{d}{dx} e^{-x} = -e^{-x} \right) (-5)$$

$$\frac{d}{dx} -5e^{-x} = 5e^{-x}$$

$$\boxed{-5e^{-x}}$$

b. $\frac{1}{3}x^{3/2}$

$$\frac{\frac{1}{3}x^{3/2+1}}{\frac{3}{2}+1} = \frac{\frac{1}{3}x^{5/2}}{5/2} = \frac{2}{5} \cdot \frac{1}{3}x^{5/2} = \boxed{\frac{2}{15}x^{5/2}}$$

c. $-3\sin(4x)$

$$\left(\frac{d}{dx} \cos 4x = -4\sin 4x \right) \frac{3}{4}$$

$$\frac{d}{dx} \frac{3}{4} \cos 4x = -3\sin 4x$$

$$\boxed{\frac{3}{4} \cos 4x}$$